Quadratics: Expanding & Factorising

Practice test 3 · 40 marks · Higher tier · non-calculator

Name: _____ Date: _____ Score: _____ / 40

Section A — Expanding double brackets

A1. Expand and simplify (x + 2)(x + 11)

[2 marks]

A2. Expand and simplify (x – 7)(x + 7). Explain why the answer has no x term.

[2 marks]

A3. Expand and simplify (4x – 3)(x + 2)

[2 marks]

A4. Expand and simplify $(x + 3)^2 + (x – 1)^2$

[2 marks]

Section B — Expanding three brackets

B1. Expand and simplify (2x – 1)(x + 3)(x – 2)

[3 marks]

B2. Show that $(x - 1)^3 = x^3 - 3x^2 + 3x - 1$

[3 marks]

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Factorise $x^2 + 15x + 56$

[2 marks]

C2. Factorise $x^2 + x - 30$

[2 marks]

C3. Factorise $100 - x^2$

[1 mark]

C4. Factorise fully $5x^2 - 20x - 60$

[2 marks]

Section D — Factorising $ax^2 + bx + c$

D1. Factorise $3x^2 + 11x + 10$

[2 marks]

D2. Factorise $4x^2 - 4x - 15$

[2 marks]

D3. Factorise $25x^2 - 9y^2$

[2 marks]

Section E — Using factorising

E1. Solve $x^2 = 6x - 8$

[3 marks]

E2. Simplify fully

$$\frac{3x^2 + 6x}{x^2 - x - 6}$$

[3 marks]

Section F — Problem solving & proof

F1. The area of a square is $(x^2 + 14x + 49)\ cm^2$. Find an expression for its perimeter.

[2 marks]

F2. A cuboid measures x cm by (x + 2) cm by (x + 5) cm. Find an expression for its volume. Give your answer in expanded form.

[2 marks]

F3. Prove that the sum of the squares of two consecutive whole numbers is always odd.

[2 marks]

F4. Without a calculator, work out $7.5^2 - 2.5^2$

[1 mark]